3. cvičení

Příklady

IES FSV UK · Josef Hlahulek

1. $\lim\limits_{n\to\infty} \dfrac{\left(1+\frac{1}{n}\right)^{120} - \left(1+\frac{2}{n}\right)^{80}}{\left(1-\frac{1}{n}\right)^{100} + \left(1+\frac{3}{n}\right)^{100} - 2}$

2. $\lim\limits_{n\to\infty} \dfrac{\left(1+\frac{1}{n}\right)^{150} - \left(1+\frac{2}{n}\right)^{60}}{\left(1-\frac{1}{n}\right)^{99} + \left(1+\frac{3}{n}\right)^{99} - 2}$

3. $\lim\limits_{n\to\infty} \dfrac{\sqrt[3]{n^3+5n} - \sqrt[3]{n^3+2}}{\sqrt{n^4+n+1} - \sqrt{n^4+1}}$

4. $\lim\limits_{n\to\infty} \dfrac{\sqrt[3]{n^3+n+1} - \sqrt[3]{n^3-n+2}}{\sqrt{n^4+2n+3} - \sqrt{n^4+n+4}}$

5. $\lim\limits_{n\to\infty} \dfrac{(n+7)^{50} - (n^2+1)^{25}}{\sqrt{n^{100}+n^{99}-1} - \sqrt{n^{100}+2n^{99}+1}}$

6. $\lim\limits_{n\to\infty} \dfrac{\sqrt[3]{n^3+6n} - \sqrt[3]{n^3+7}}{\sqrt{n^2+4} - \sqrt{n^2+1}}$

7. $\lim\limits_{n\to\infty} \dfrac{\sqrt[n]{2^n+3^n}}{\sqrt[2n]{4^n+\sqrt{n}}}$

8. $\lim\limits_{n\to\infty} \sqrt[n]{n^2+n^3+n^4+2^n+3^n+4^n}$

9. $\lim\limits_{n\to\infty} \dfrac{(n^2+1)^{100} - (n+2)^{200} + 400n^{199}}{1+2+3+\ldots+n^{99}}$

Výsledky

1. $-\frac{1}{5}$ 2. $\frac{5}{33}$ 3. $\frac{10}{3}$ 4. $\frac{4}{3}$ 5. $+\infty$ 6. $\frac{4}{3}$ 7. $\frac{3}{2}$ 8. 4 9. -159000